ENVS 423L/EPS 522 Problem Set #2

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Calculations, methods, and assumptions should be spelled out so that your work can be reproduced. All work should be shown to receive full credit.

Upload a word document or PDF of your solutions/answers by midnight on Thursday, September 4 to Canvas.

1 Introduction

The purpose of this problem set is to reinforce your understanding of the climate system.

1.1 Black Body Emission

Estimate the emission flux for each of the following, assuming they are a black body (10 pts):

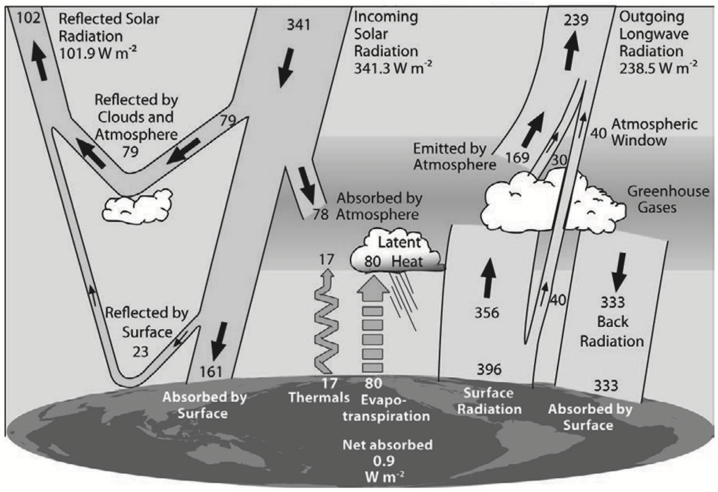
Surface	Surface	Snow	Cloud	Sun
Temperature (°C)	15	0	–20	6000

Wien's displacement law states that the black-body radiation curve for different temperatures will peak at different wavelengths inversely proportional to the temperature:

$$\lambda_{peak} = \frac{b}{T} \tag{1}$$

where b is Wien's displacement constant (2898 μm) and T is the absolute temperature (K). What is the peak wavelength for each of the above and what part of the spectrum do those correspond to? (10 pts)

1.2 Albedo



Based on the above energy budget graphic, what is the average albedo of Earth's atmosphere? (5 pts)

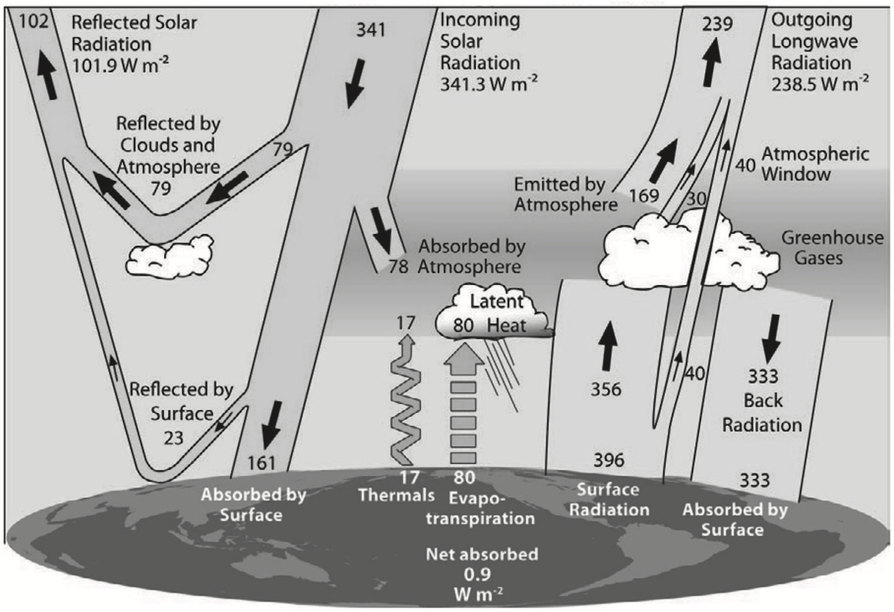
What is the average albedo of Earth's surface? (5 pts)

Compare the two values above and give a reason for the difference. (5 pts)

How might climate change affect earth surface albedo and, therefore, the energy balance? (5 pts)

1.3 Energy Balance

In class we conducted an energy balance at the top of the atmosphere and at Earth's surface. Do the same for the atmosphere. Is it perfectly balanced based on the numbers in this figure, and if not, why? (10 pts)



1.4 Top of Atmosphere Shortwave Radiation

Estimate the top of atmosphere (TOA) shortwave (SW) radiation in Albuquerque, NM on our first day of class (August 19) assuming "solar noon." (10 pts)

Now, estimate and plot the TOA SW radiation throughout the calendar year for Albuquerque at solar noon. There is a dataset called avg_daily_temp_abq.csv that contains the daily average temperature for Albuquerque. Add that to the plot and discuss why the two curves don't perfectly line up. (10 pts)

1.5 General Circulation

Hand draw Earth's general circulation, labeling major cells (and their names), zones of high low/pressure, areas of higher and lower precipitation, wind direction, and temperature of circulating air masses. (15 pts)

1.6 New Mexico Climate

What is the Koppen-Geiger climate classification for Albuquerque, NM? (5 pts)

Explain, using concepts from the climate system lecture, how this climate arises. (10 pts)